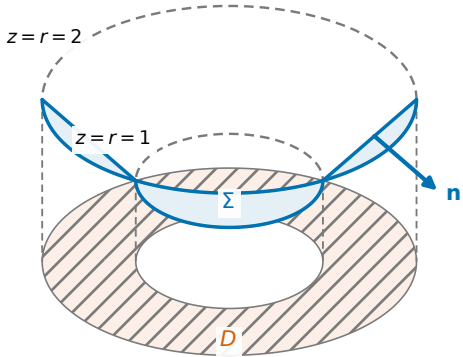
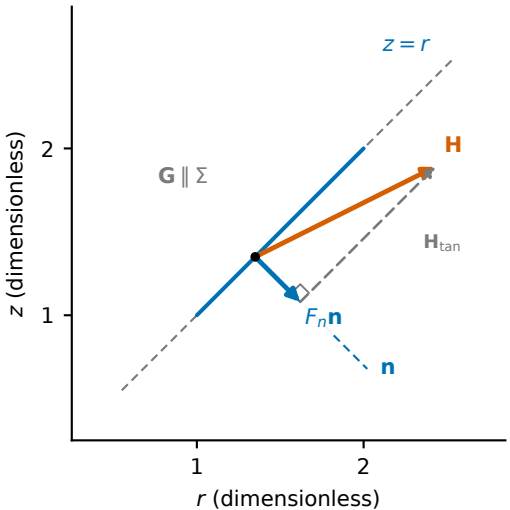
A



$$\mathbf{n} = \tfrac{1}{\sqrt{2}}(x/r,\ y/r,\ -1)$$

B



$$F_n = r/\sqrt{2}\,, \quad dS = \sqrt{2}\,dA_{xy}$$